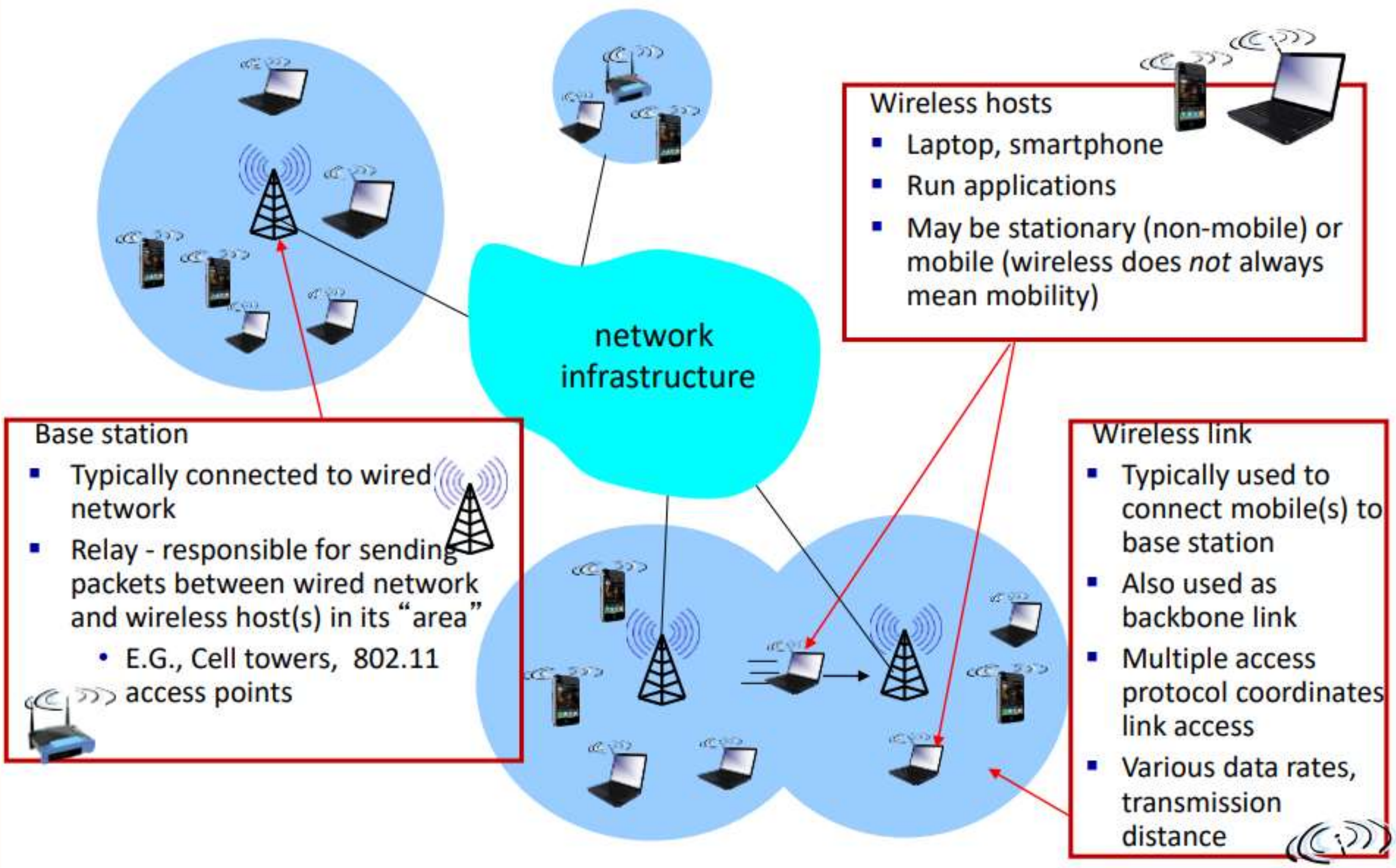
UNIVERSITY OF WOLVERHAMPTON

Wireless Networks

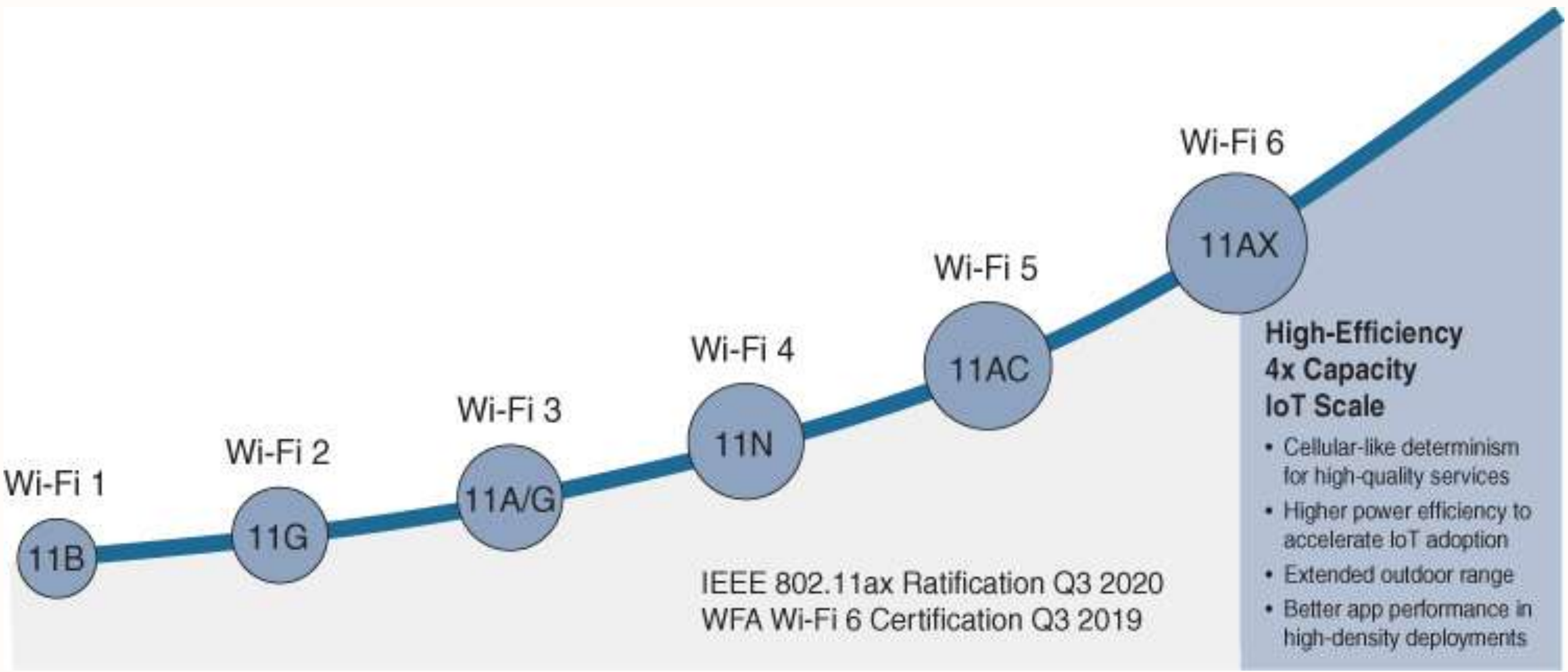
6CS029 Advanced Networking

Elements of a wireless network



Types of Wireless Networks

- **Wireless Personal-Area Network (WPAN)** – Low power and short-range (20-30ft or 6-9 meters). Based on IEEE 802.15 standard and 2.4 GHz frequency. Bluetooth and Zigbee are WPAN examples.
- **Wireless LAN (WLAN)** – Medium sized networks up to about 300 feet. Based on IEEE 802.11 standard and 2.4 or 5.0 GHz frequency.
- **Wireless MAN (WMAN)** and **Wireless WAN (WWAN)** – Large geographic area such as city or district or Extensive geographic area for national or global communication
    - Uses specific licensed frequencies.
    - **WiMAX (Worldwide Interoperability for Microwave Access)** – Alternative broadband wired internet connections. IEEE 802.16 WLAN standard for up 30 miles (50 km).
    - **Cellular Broadband** – Carry both voice and data. Used by phones, automobiles, tablets, and laptops.
    - **Satellite Broadband** – Uses directional satellite dish aligned with satellite in geostationary orbit. Needs clear line of site. Typically used in rural locations where cable and DSL are unavailable.


Wi-Fi 1
11B
Wi-Fi 2
11G
Wi-Fi 3
11A/G
Wi-Fi 4
11N
Wi-Fi 5
11AC
Wi-Fi 6
11AX
IEEE 802.11ax Ratification Q3 2020
WFA Wi-Fi 6 Certification Q3 2019
High-Efficiency
4x Capacity
IoT Scale
• Cellular-like determinism for high-quality services
• Higher power efficiency to accelerate IoT adoption
• Extended outdoor range
• Better app performance in high-density deployments

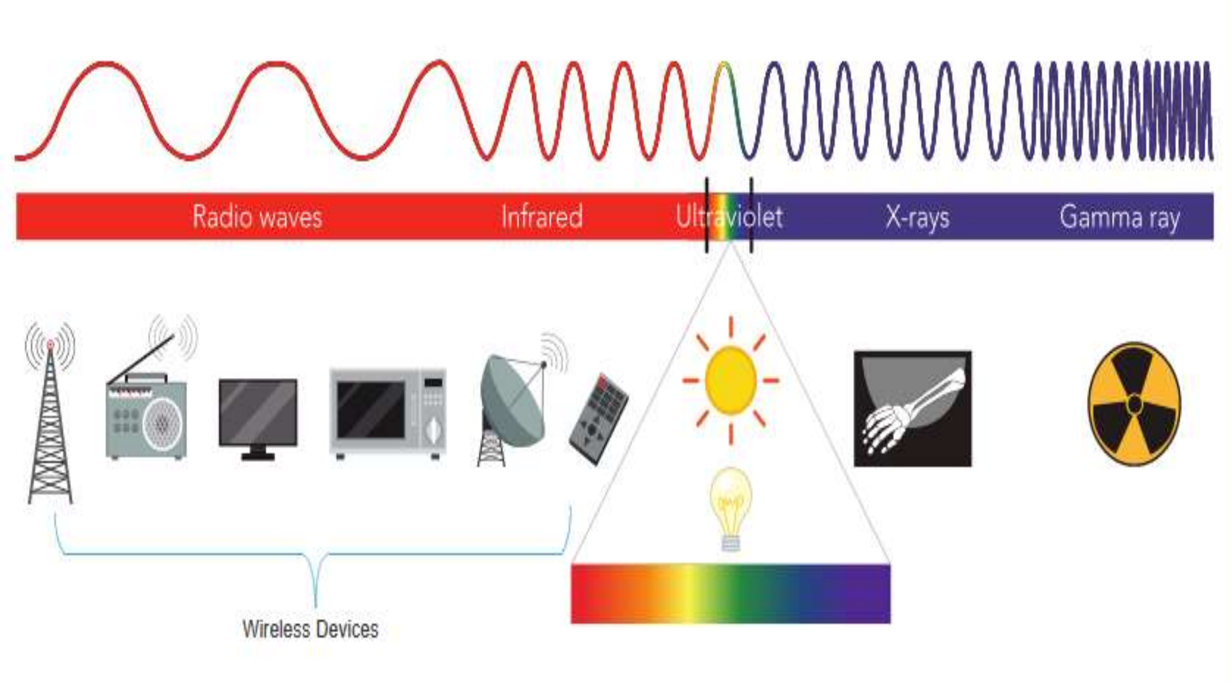
802.11 Standards

IEEE Standard	Radio Frequency	Description
802.11	2.4 GHz	Data rates up to 2 Mb/s
802.11a	5 GHz	Data rates up to 54 Mb/s Not interoperable with 802.11b or 802.11g
802.11b	2.4 GHz	Data rates up to 11 Mb/s Longer range than 802.11a and better able to penetrate building structures
802.11g	2.4 GHz	Data rates up to 54 Mb/s Backward compatible with 802.11b
802.11n	2.4 and 5 GHz	Data rates 150 – 600 Mb/s Require multiple antennas with MIMO technology
802.11ac	5 GHz	Data rates 450 Mb/s – 1.3 Gb/s Supports up to eight antennas
802.11ax	2.4 and 5 GHz	High-Efficiency Wireless (HEW) Capable of using 1 GHz and 7 GHz frequencies

Radio Frequencies

All wireless devices operate in the range of the electromagnetic spectrum. WLAN networks operate in the 2.4 and 5 GHz frequency bands.

- 2.4 GHz (UHF) – 802.11b/g/n/ax
- 5 GHz (SHF) – 802.11a/n/ac/ax



WLAN Components

Wireless Home Router

A home user typically interconnects wireless devices using a small, wireless router.

Wireless routers serve as the following:

–**Access point** – To provide wires access

–**Switch** – To interconnect wired devices

–**Router** - To provide a default gateway to other networks and the Internet

Wireless Access Point

- Wireless clients use their wireless NIC to discover nearby access points (APs).
- Clients then attempt to associate and authenticate with an AP.
- After being authenticated, wireless users have access to network resources.

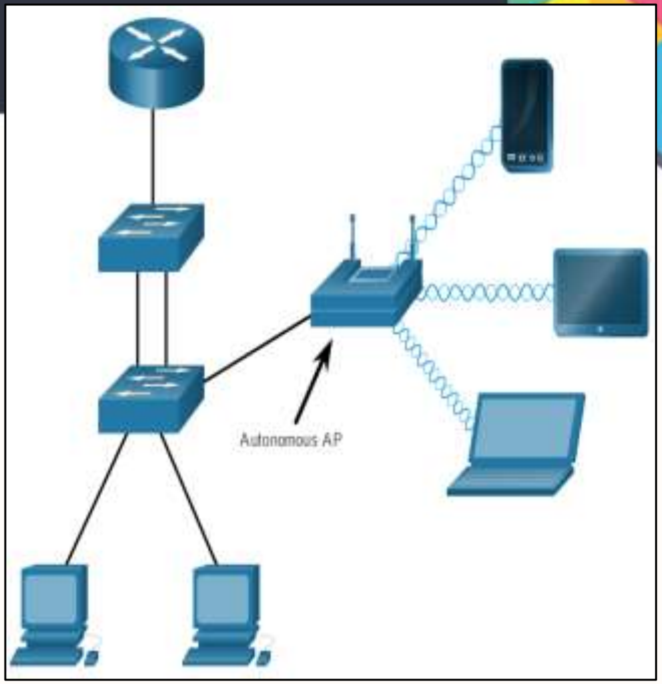
Cisco Meraki Go access points

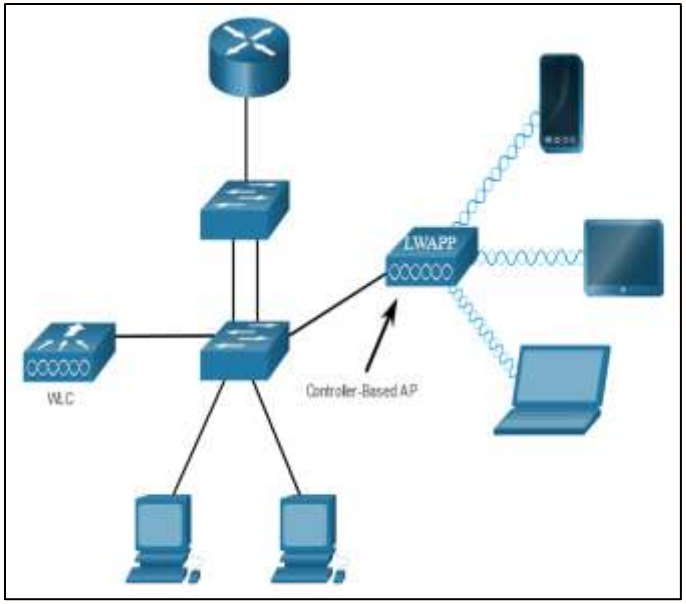
Types of external antennas:

- **Omnidirectional** – Provide 360-degree coverage. Ideal in houses and office areas.
- **Directional** – Focus the radio signal in a specific direction. Examples are the Yagi and parabolic dish.
- **Multiple Input Multiple Output (MIMO)** – Uses multiple antennas (Up to eight) to increase bandwidth.

AP Categories

- **Autonomous APs** – Standalone devices. Each autonomous AP acts independently of the others and is configured and managed manually by an administrator.
- **Controller-based APs** – Also known as lightweight APs (LAPs). Use Lightweight Access Point Protocol (LWAPP) to communicate with a LWAN controller (WLC). Each LAP is automatically configured and managed by the WLC.

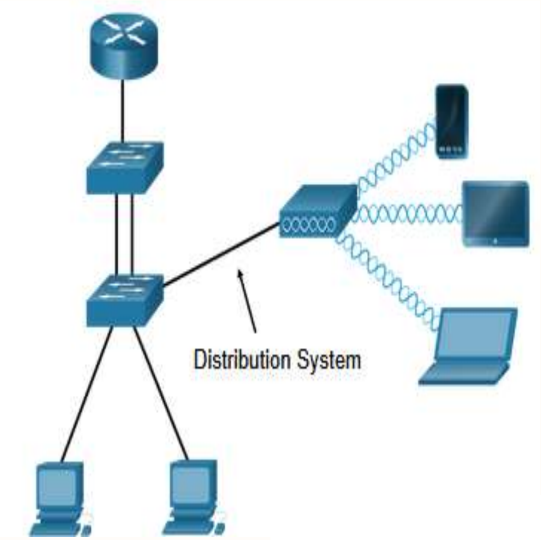




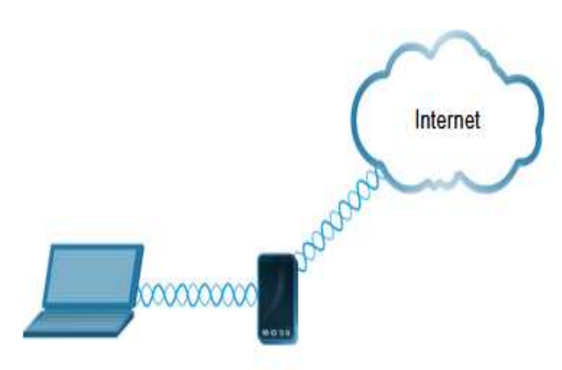
WLAN Operation

802.11 Wireless Topology Modes

Ad hoc mode - Used to connect clients in peer-to-peer manner without an AP.



Infrastructure mode - Used to connect clients to the network using an AP.



Tethering - Variation of the ad hoc topology is when a smart phone or tablet with cellular data access is enabled to create a personal hotspot.

BSS and ESS

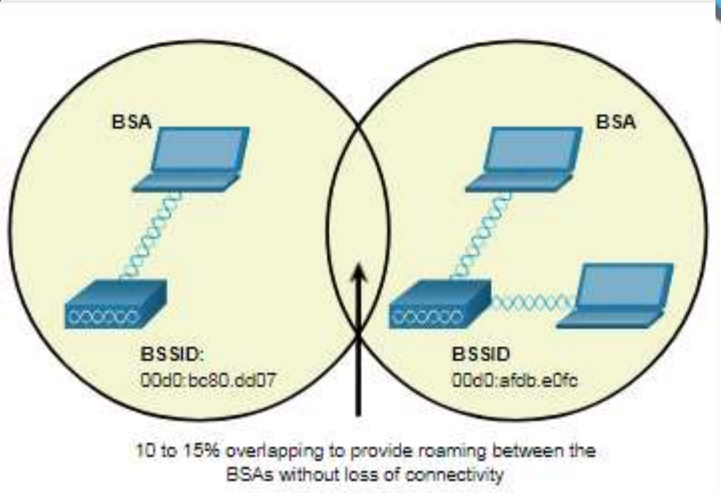
Infrastructure mode defines two topology blocks:

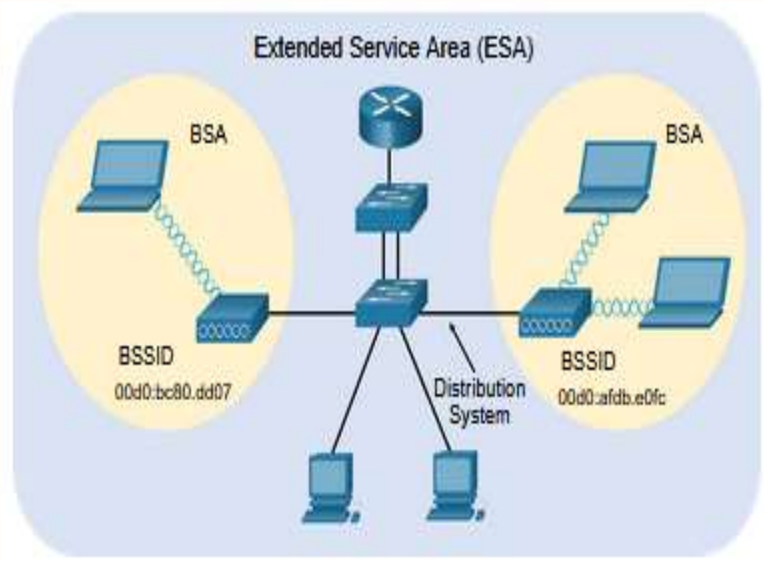
Basic Service Set (BSS)

- Uses single AP to interconnect all associated wireless clients.
- Clients in different BSSs cannot communicate.

Extended Service Set (ESS)

- A union of two or more BSSs interconnected by a wired distribution system.
- Clients in each BSS can communication through the ESS.





Transition Types Based On Mobility

- No transition
  - Stationary or moves only within BSS
- BSS transition
  - Station moving from one BSS to another BSS in same ESS
- ESS transition
  - Station moving from BSS in one ESS to BSS within another ESS

Wireless Link Characteristics

Important differences from wired link ….

- Decreased signal strength: radio signal attenuates as it propagates through matter (path loss)
- Interference from other sources: standardized wireless network frequencies (e.g., 2.4 ghz) shared by other devices (e.g., Phone); devices (motors) interfere as well
- Multipath propagation: radio signal reflects off objects ground, arriving at destination at slightly different times

Make communication across (even a point to point) wireless link much more “difficult”

Medium Access Control

MAC layer covers three functional areas:

- Reliable data delivery
  - More efficient to deal with errors at the MAC level than higher layer (such as TCP)
  - Frame exchange protocol
    - Source station transmits data
    - Destination responds with acknowledgment (ACK)
    - If source doesn‘t receive ACK, it retransmits frame
- Access control
  - Avoiding collisions
  - Carrier Sense Multiple Access/ Collision Avoidance (CSMA/CA)
- Security

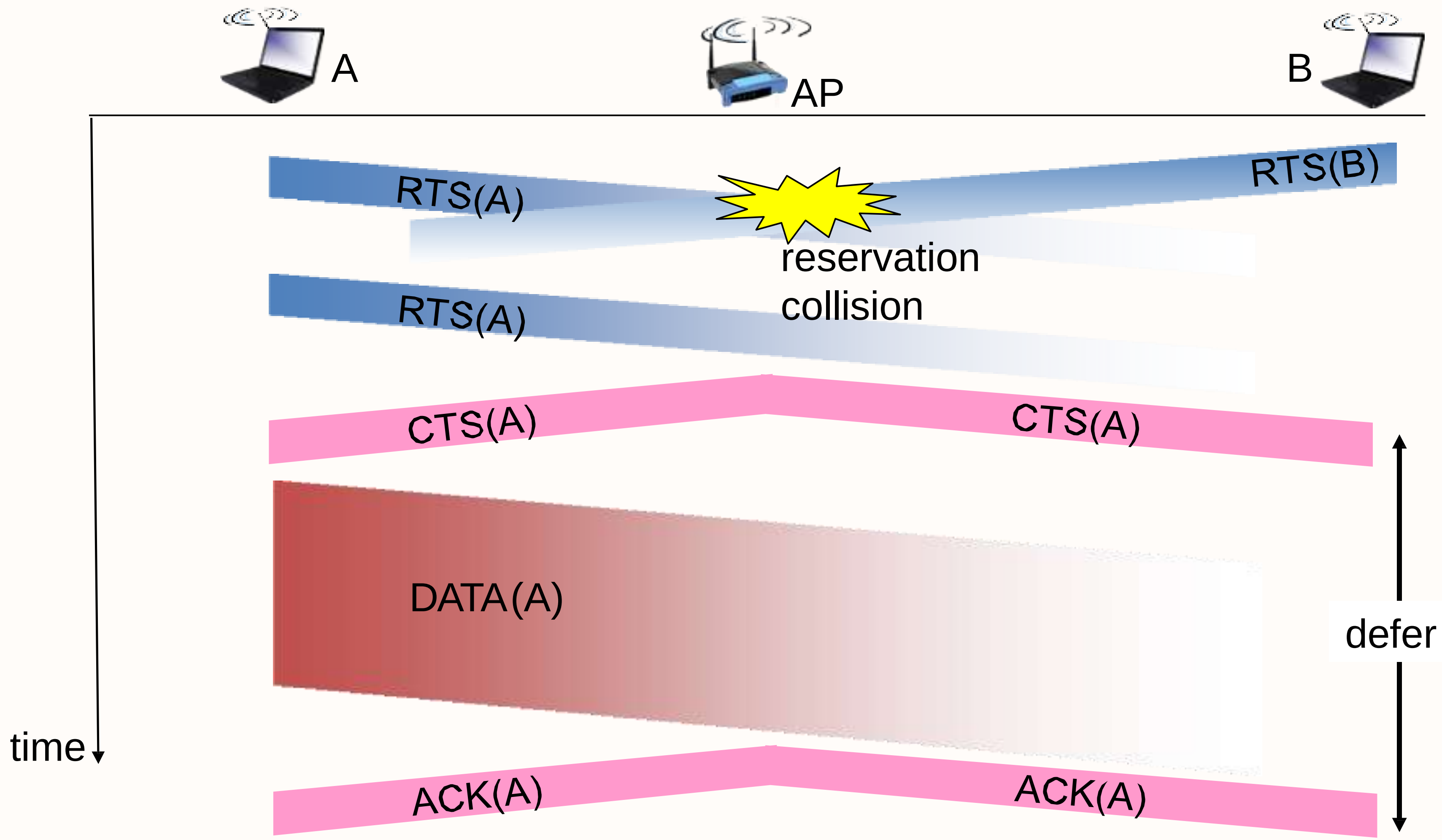
CSMA/CA

WLANs are half-duplex and a client cannot “hear” while it is sending, making it impossible to detect a collision.

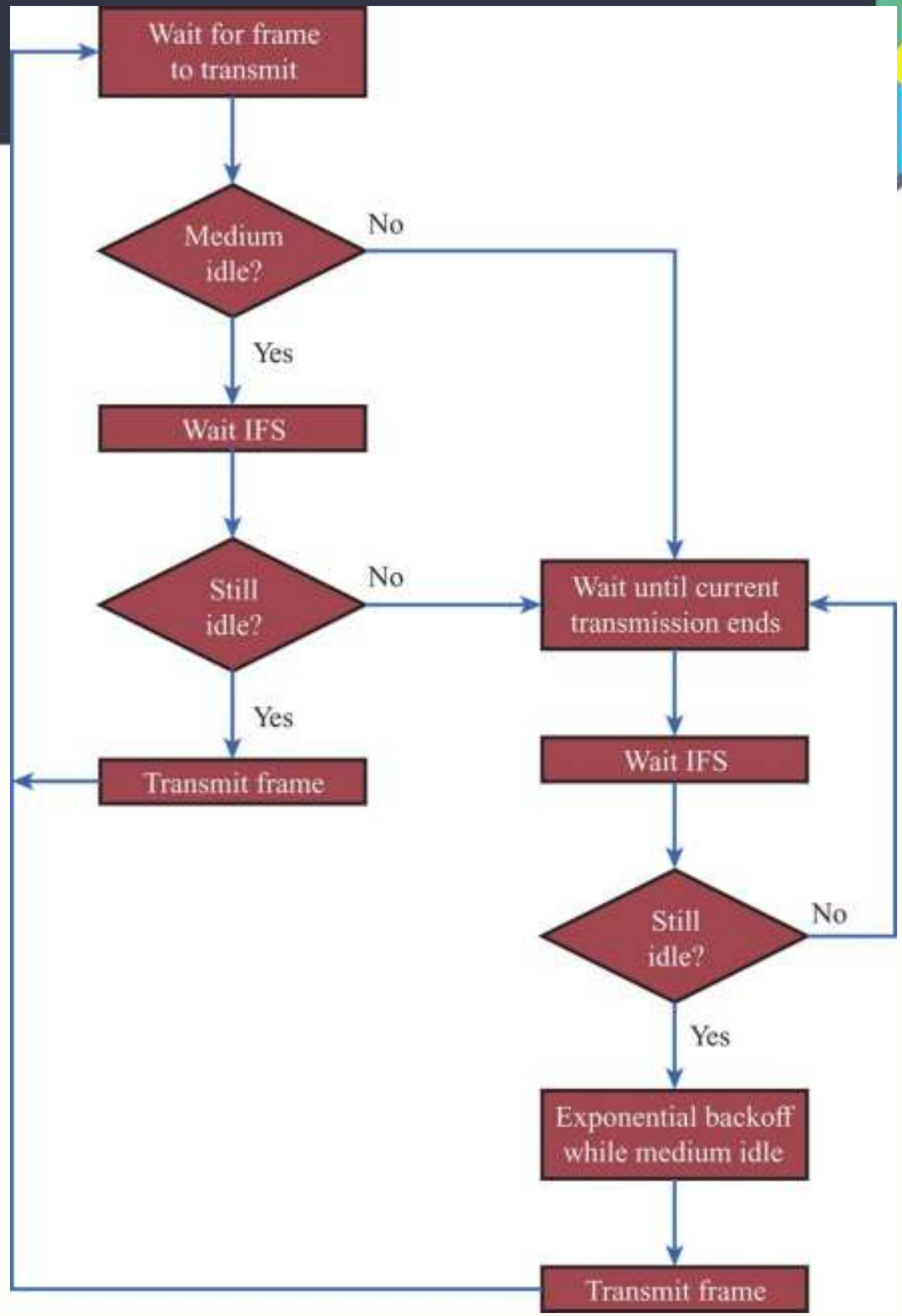
WLANs use carrier sense multiple access with collision avoidance (CSMA/CA) to determine how and when to send data. A wireless client does the following:

1. Listens to the channel to see if it is idle, i.e. no other traffic currently on the channel.
2. Sends a ready to send (RTS) message the AP to request dedicated access to the network.
3. Receives a clear to send (CTS) message from the AP granting access to send.
4. Waits a random amount of time before restarting the process if no CTS message received.
5. Transmits the data.
6. Acknowledges all transmissions. If a wireless client does not receive an acknowledgment, it assumes a collision occurred and restarts the process

Collision Avoidance: RTS-CTS exchange



IEEE 802.11 Medium Access Control Logic

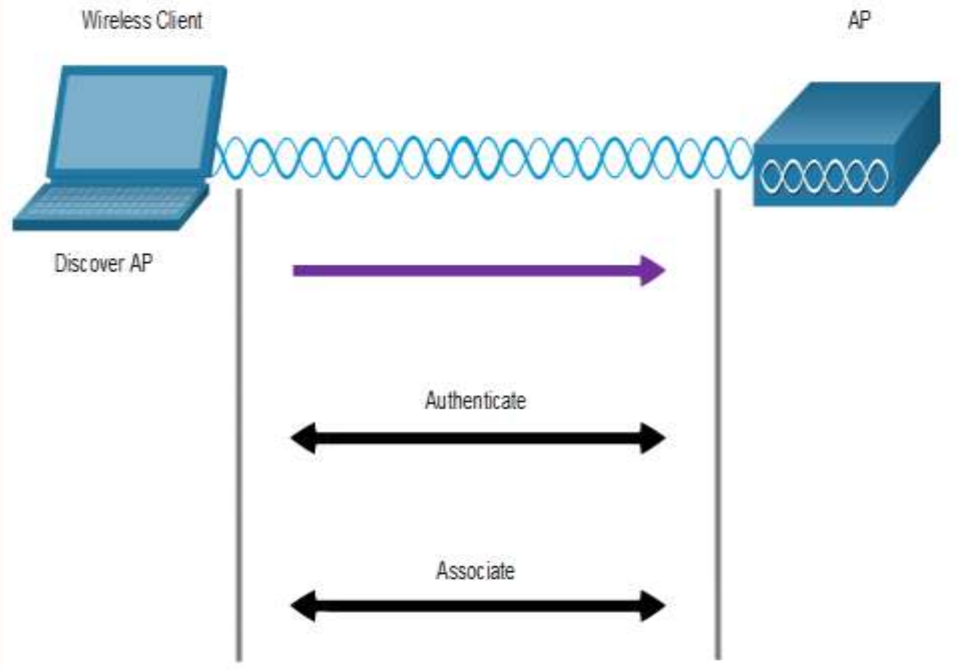


Wireless Client and AP Association

Wireless devices must associate with an AP or wireless router.

Wireless devices complete the following three stage process:

- Discover a wireless AP
- Authenticate with the AP
- Associate with the AP



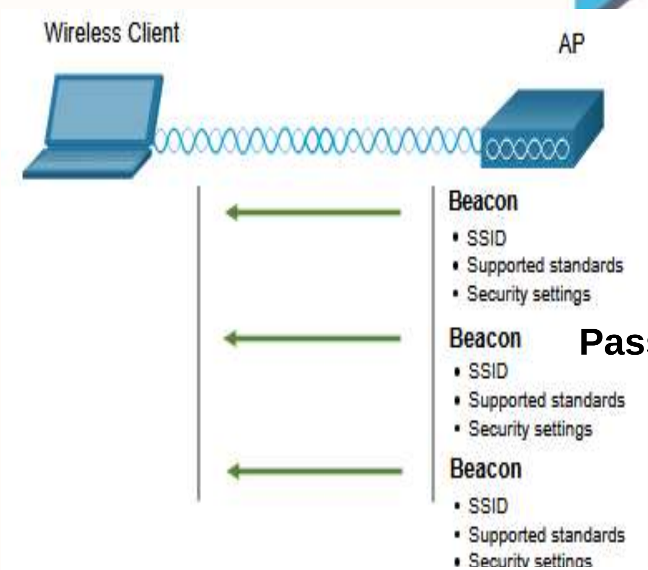
A wireless client and an AP must agree on specific parameters:

SSID – The client needs to know the name of the network to connect.
Password – This is required for the client to authenticate to the AP.
Network mode – The 802.11 standard in use.
Security mode – The security parameter settings, i.e. WEP, WPA, or WPA2.
Channel settings – The frequency bands in use.

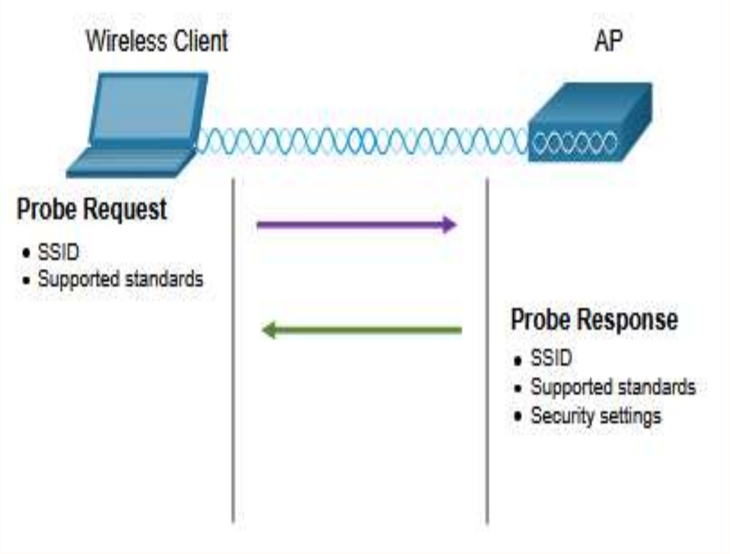
Discover Mode

Wireless clients connect to the AP using a passive or active scanning (probing) process.

- **Passive mode** – AP openly advertises its service by periodically sending broadcast beacon frames containing the SSID, supported standards, and security settings.
- **Active mode** – Wireless clients must know the name of the SSID. The wireless client initiates the process by broadcasting a probe request frame on multiple channels.



Passive mode



Active mode

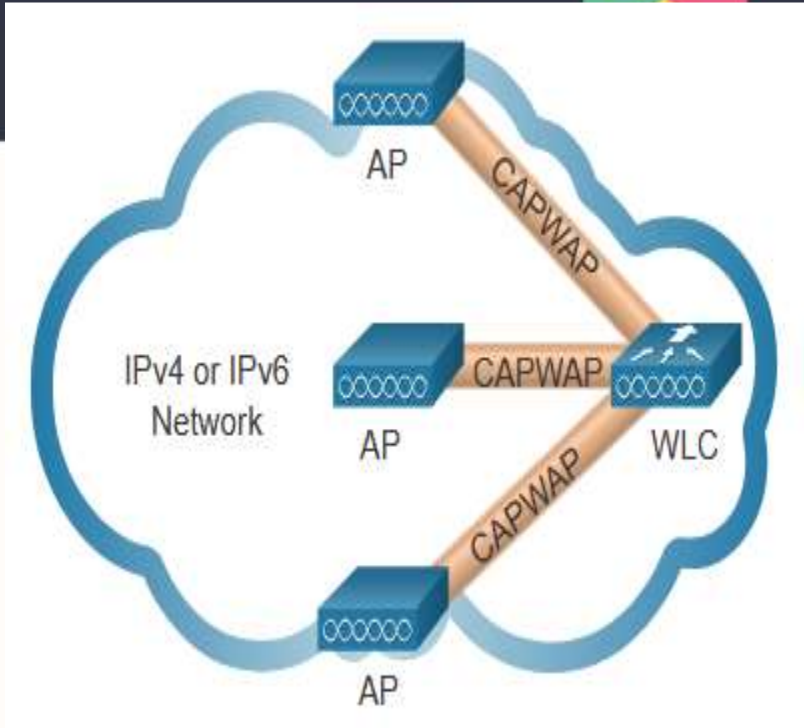
Association-Related Services

- Association
  - Establishes initial association between station and AP
- Reassociation
  - Enables transfer of association from one AP to another, allowing station to move from one BSS to another
- Disassociation
  - Association termination notice from station or AP

Control and Provisioning of Wireless Access Points (CAPWAP)

CAPWAP

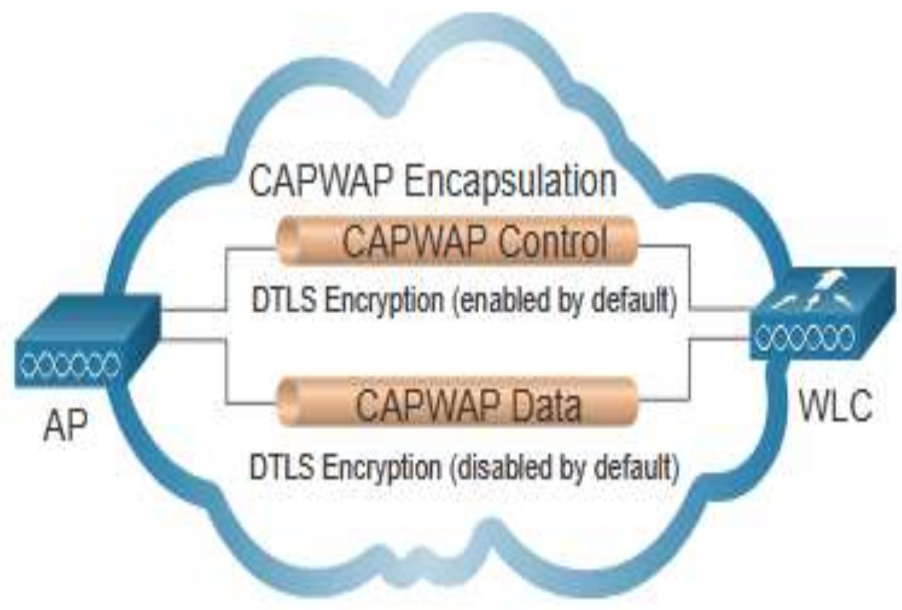
- IEEE standard protocol that enables a WLC to manage multiple APs and WLANs.
- Based on LWAPP but adds additional security with Datagram Transport Layer Security (DLTS).
- Encapsulates and forwards WLAN client traffic between an AP and a WLC over tunnels using UDP ports 5246 and 5247.
- Operates over both IPv4 and IPv6. IPv4 uses IP protocol 17 and IPv6 uses IP protocol 136.
- MAC functions are split between WLC and APs



AP MAC Functions	WLC MAC Functions
Beacons and probe responses	Authentication
Packet acknowledgements and retransmissions	Association and re-association of roaming clients
Frame queueing and packet prioritization	Frame translation to other protocols
MAC layer data encryption and decryption	Termination of 802.11 traffic on a wired interface

DTLS Encryption

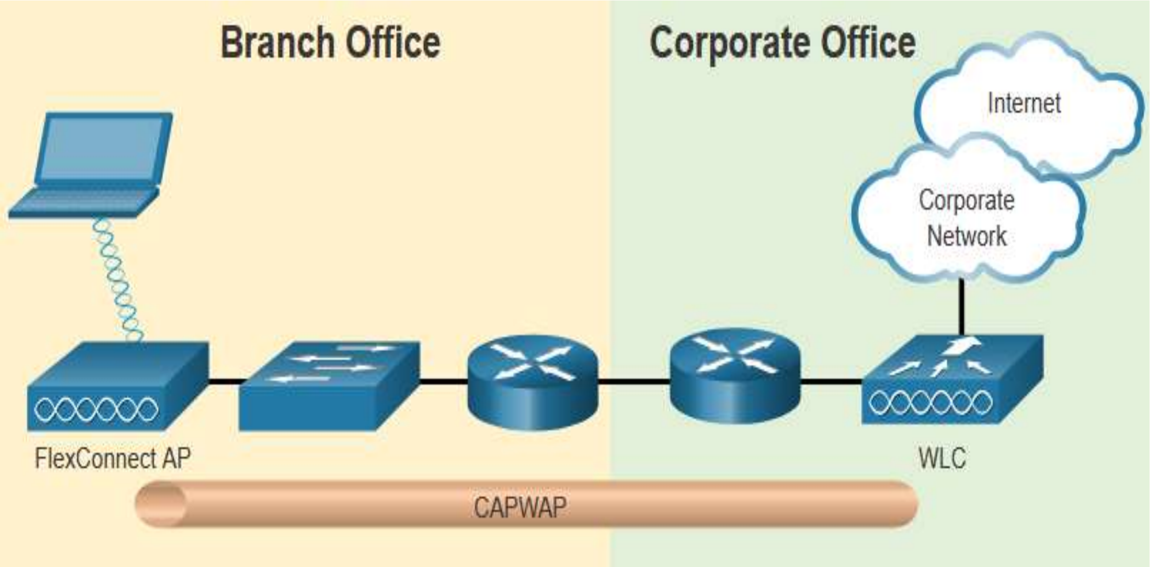
- DTLS provides security between the AP and the WLC.
- It is enabled by default to secure the CAPWAP control channel and encrypt all management and control traffic between AP and WLC.
- Data encryption is disabled by default and requires a DTLS license to be installed on the WLC before it can be enabled on the AP.



Flex Connect APs

FlexConnect enables the configuration and control of Aps over a WAN link.

- **Connected mode** – The WLC is reachable. The FlexConnect AP has CAPWAP connectivity with the WLC through the CAPWAP tunnel. The WLC performs all CAPWAP functions.
- **Standalone mode** – The WLC is unreachable. The FlexConnect AP has lost CAPWAP connectivity with the WLC. The FlexConnect AP can assume some of the WLC functions such as switching client data traffic locally and performing client authentication locally.

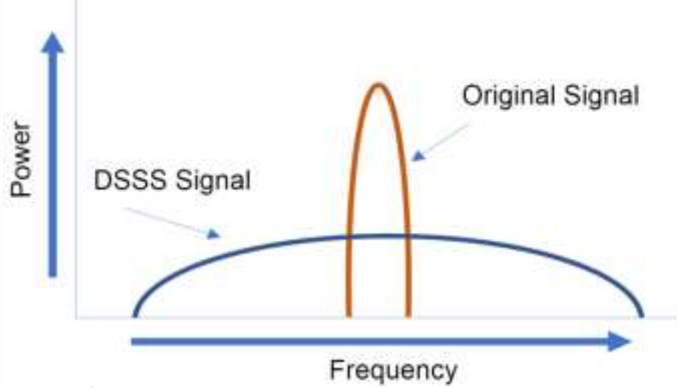


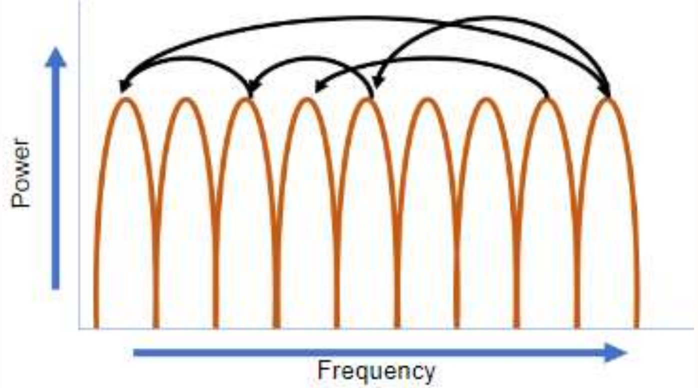
Channel Management

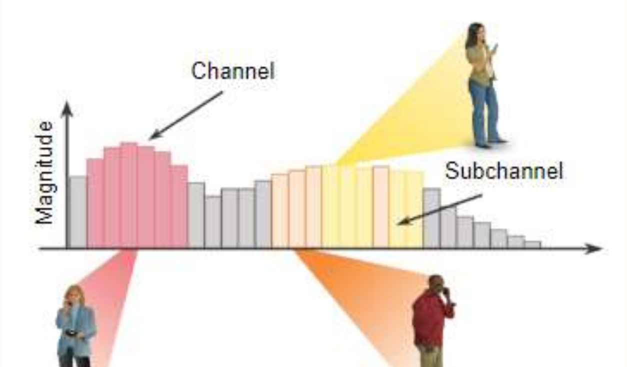
Frequency Channel Saturation

Channel saturation can be mitigated using techniques that use the channels more efficiently.

- **Direct-Sequence Spread Spectrum (DSSS)** - A modulation technique designed to spread a signal over a larger frequency band.
- **Frequency-Hopping Spread Spectrum (FHSS)** - Transmits radio signals by rapidly switching a carrier signal among many frequency channels. Sender and receiver must be synchronized to “know” which channel to jump to. Used by the original 802.11 standard.
- **Orthogonal Frequency-Division Multiplexing (OFDM)** - A subset of frequency division multiplexing in which a single channel uses multiple sub-channels on adjacent frequencies. OFDM is used by a number of communication systems including 802.11a/g/n/ac.

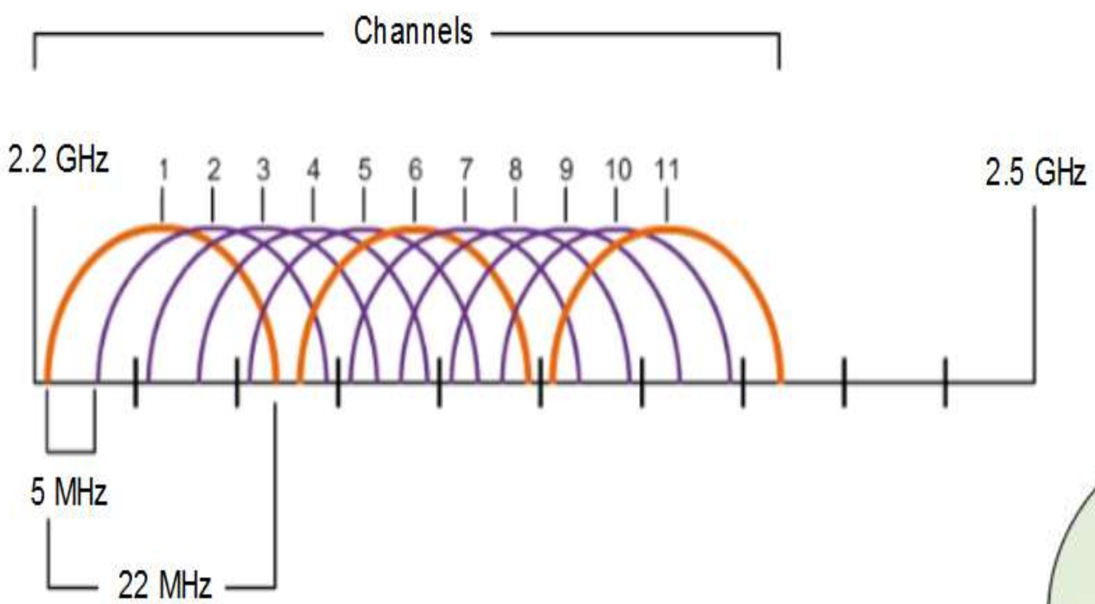


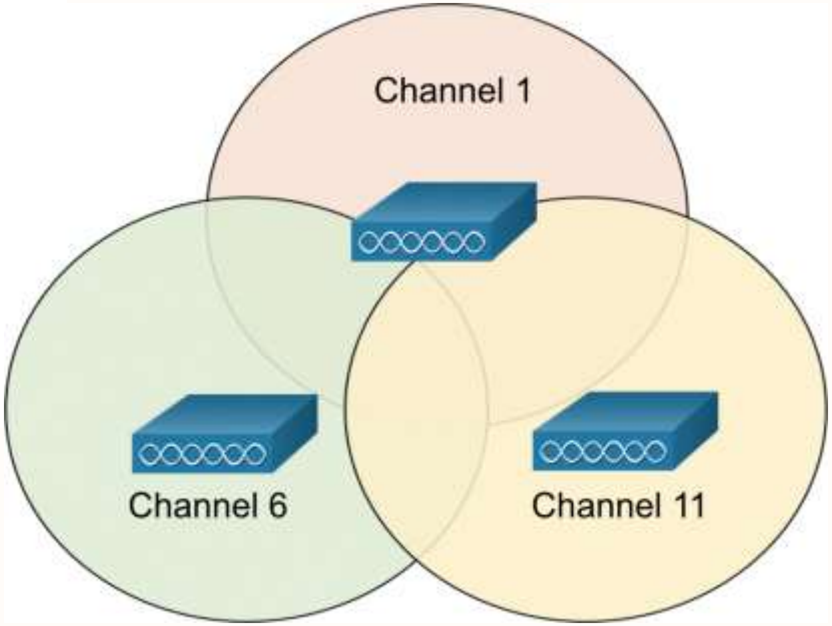




Channel Selection

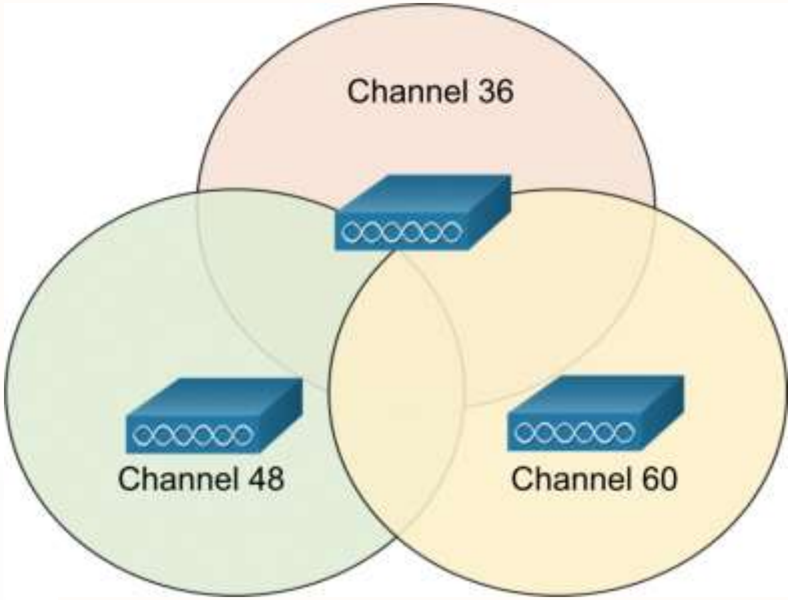
- The 2.4 GHz band is subdivided into multiple channels each allotted 22 MHz bandwidth and separated from the next channel by 5 MHz.
- A best practice for 802.11b/g/n WLANs requiring multiple APs is to use non-overlapping channels such as 1, 6, and 11.

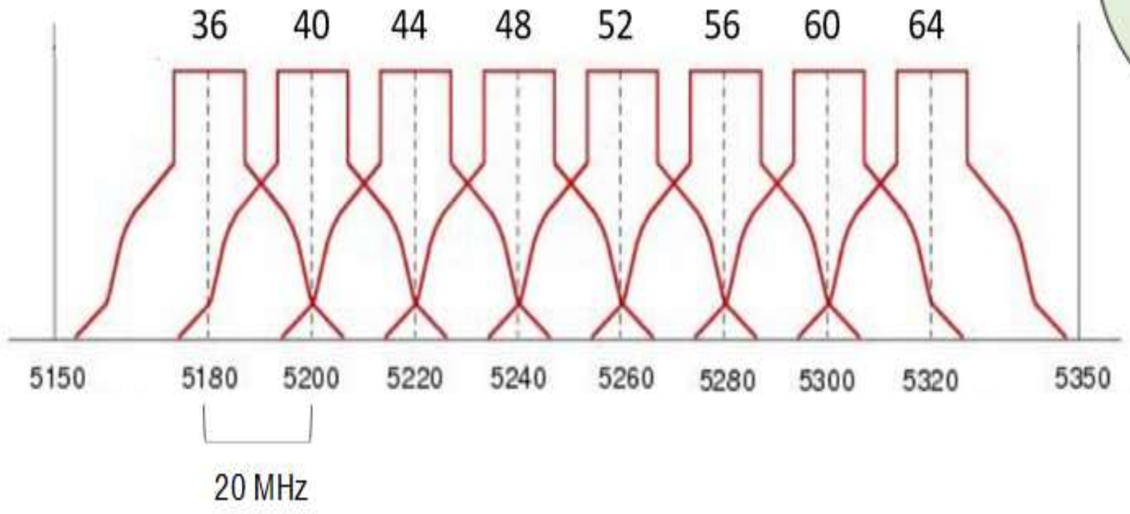




Channel Selection

- For the 5GHz standards 802.11a/n/ac, there are 24 channels. Each channel is separated from the next channel by 20 MHz.
- Non-overlapping channels are 36, 48, and 60.



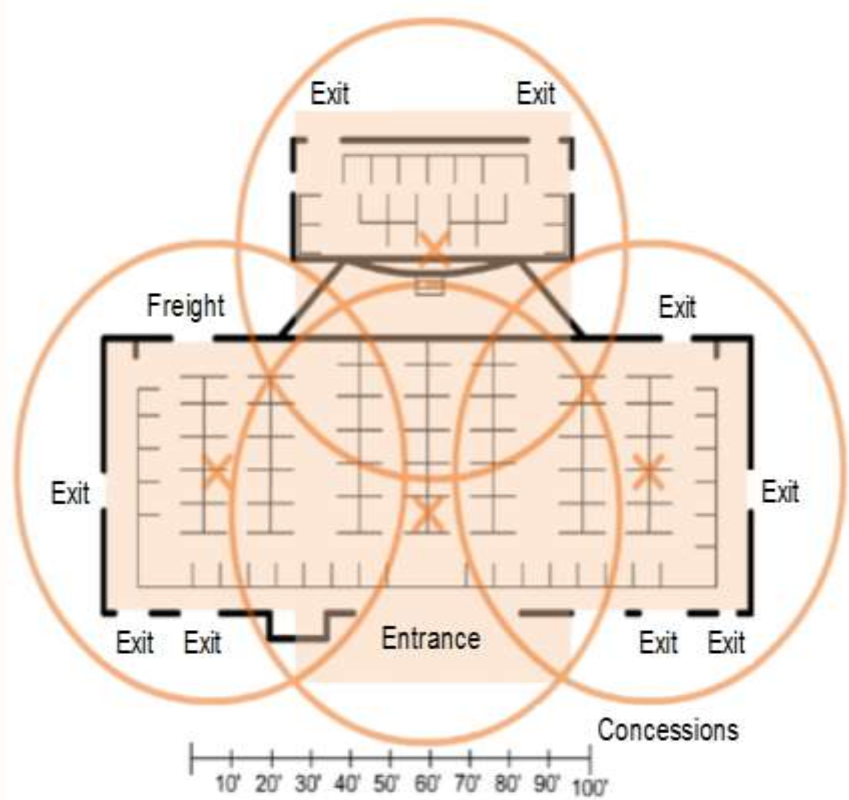


Plan a WLAN Deployment

The number of users supported by a WLAN depends on the following:

- The geographical layout of the facility
- The number of bodies and devices that can fit in a space
- The data rates users expect
- The use of non-overlapping channels by multiple APs and transmit power settings

When planning the location of APs, the approximate circular coverage area is important.



Wireless Security

A WLAN is open to anyone within range of an AP and the appropriate credentials to associate to it.

Wireless networks are specifically susceptible to several threats, including the following:

- Interception of data
- Wireless intruders
- Denial of Service (DoS) Attacks
- Rogue Aps

Secure WLANs

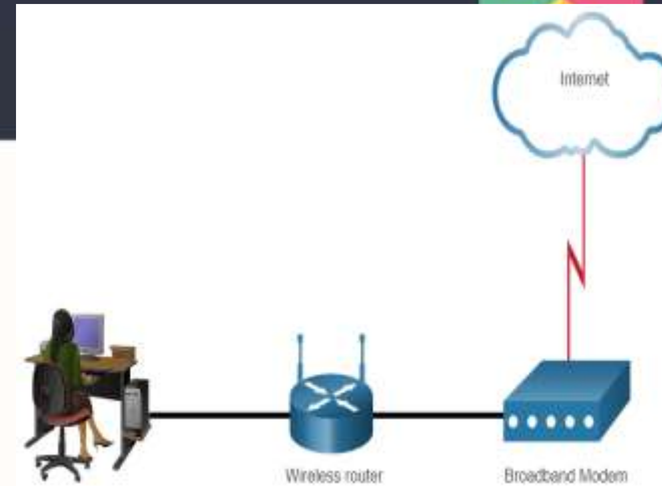
- SSID cloaking
- MAC addresses filtering
- Authentication and Encryption Systems

WLAN Configuration

The Wireless Router

Remote workers, small branch offices, and home networks often use a small office and home router.

- These “integrated” routers typically include a switch for wired clients, a port for an internet connection (sometimes labeled “WAN”), and wireless components for wireless client access.
- These wireless routers typically provide WLAN security, DHCP services, integrated Name Address Translation (NAT), quality of service (QoS), as well as a variety of other features.
- The feature set will vary based on the router model.



Basic Network Setup

Basic network setup includes the following steps:

- Log in to the router from a web browser.
- Change the default administrative password.
- Log in with the new administrative password.
- Change the default DHCP IPv4 addresses.
- Renew the IP address.
- Log in to the router with the new IP address.

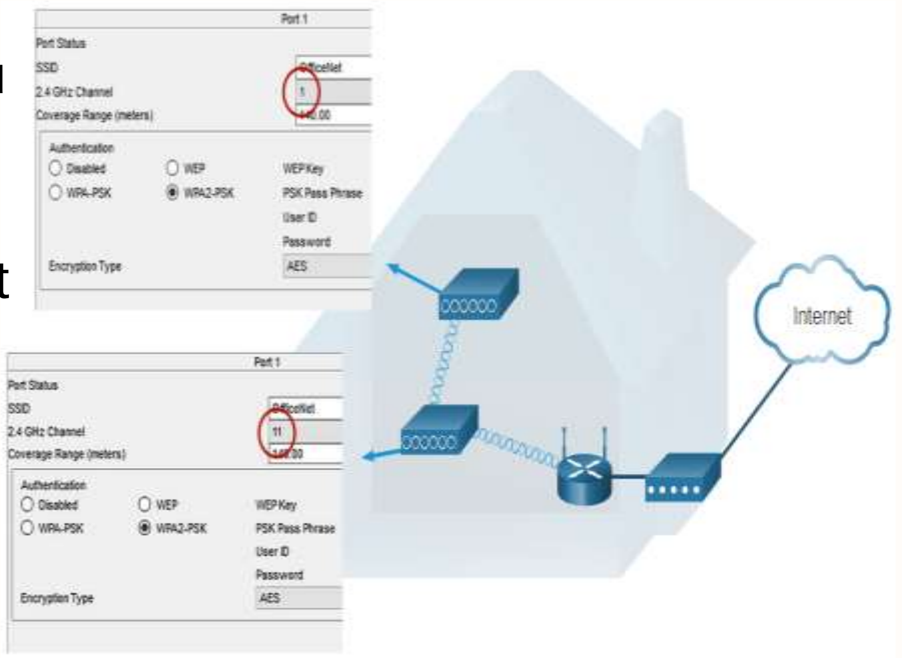
Basic wireless setup includes the following steps:

- View the WLAN defaults.
- Change the network mode, identifying which 802.11 standard is to be implemented.
- Configure the SSID.
- Configure the channel, ensuring there are no overlapping channels in use.
- Configure the security mode, selecting from Open, WPA, WPA2 Personal, WPA2 Enterprise, etc..
- Configure the passphrase, as required for the selected security mode.

Wireless Mesh Network

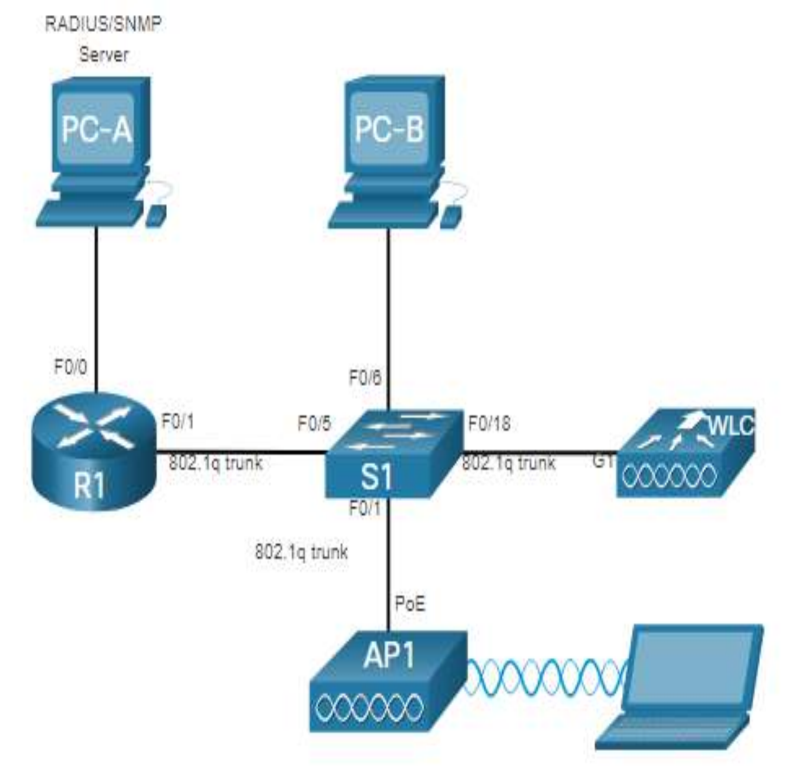
In a small office or home network, one wireless router may suffice to provide wireless access to all the clients.

- If you want to extend the range beyond approximately 45 meters indoors and 90 meters outdoors, you create a wireless mesh.
- Create the mesh by adding access points with the same settings, except using different channels to prevent interference.
- Extending a WLAN in a small office or home has become increasingly easier.
- Manufacturers have made creating a wireless mesh network (WMN) simple through smartphone apps.



WLC Topology

- The access point (AP) is a controller-based AP as opposed to an autonomous AP, so it requires no initial configuration and is often called lightweight APs (LAPs).
- LAPs use the Lightweight Access Point Protocol (LWAPP) to communicate with a WLAN controller (WLC).
- Controller-based APs are useful in situations where many APs are required in the network.
- As more APs are added, each AP is automatically configured and managed by the WLC.



Questions?